Oxford College of Emory University

Money Creation and Central Banks

Guilherme Paiva Pinto

October 29, 2026

Roadmap

1. Explain how banks create money.
2. Calculate and interpret the simple money multiplier.
3. Understand how central banks work and identify their policy tools.
4. Understand how lessons from financial crises shaped monetary policymaking.

Chapter 14, Part 2 · Section 1

Explain how banks create money.

Trace deposits, loans and payments, distinguishing changes in the money supply from changes in the monetary base.

Determining the Money Supply

Money supply = public currency + checkable deposits

Simplified measure: M = C + D

The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing			

Determining the Money Supply

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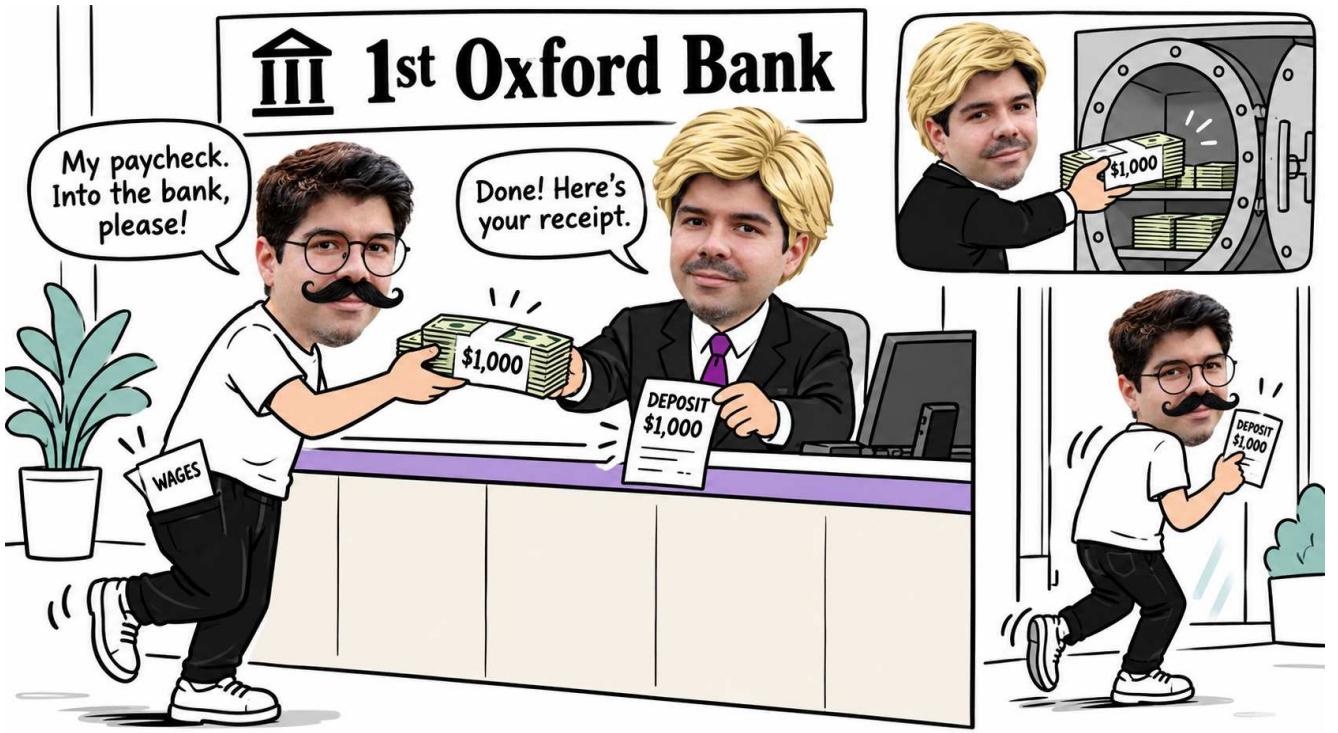
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The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing	$0	$1,000	$1,000

The deposit changes the form of money, not its total.

How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%



The cash deposit changes the form of $1,000, not the money total.

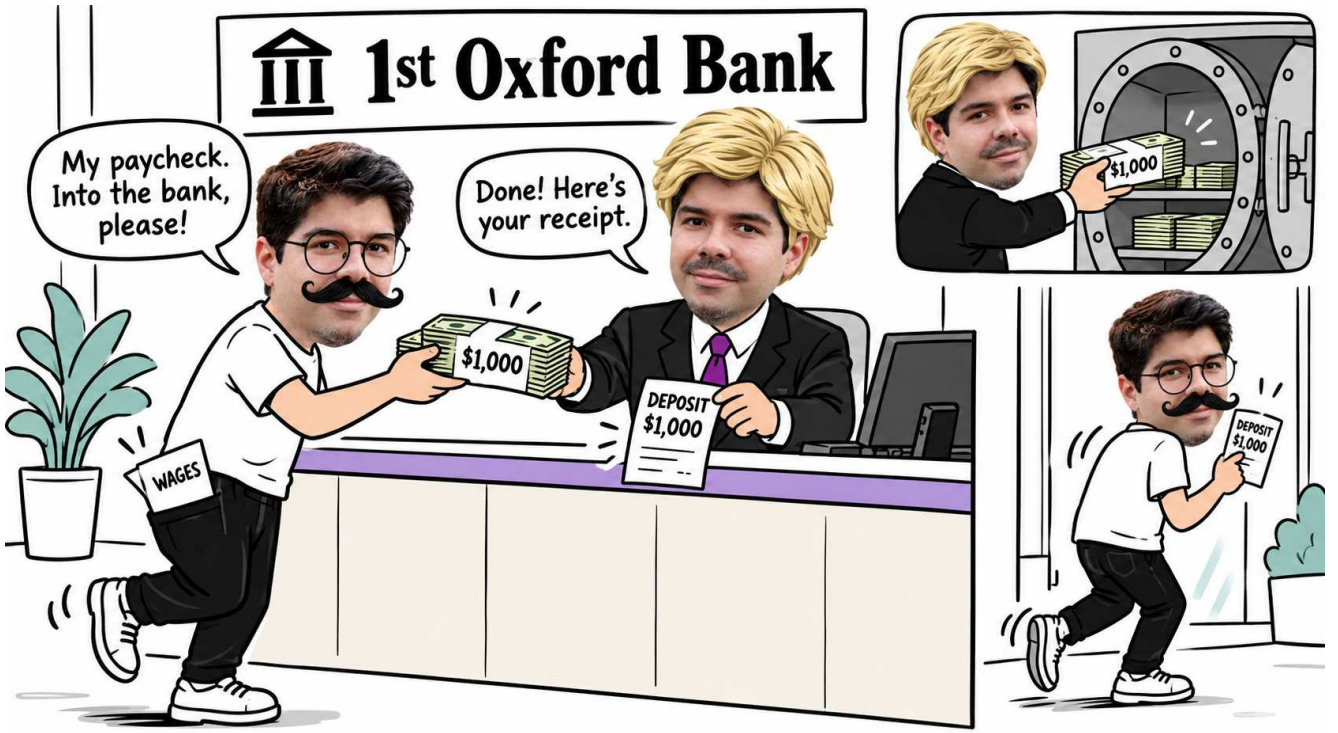
How banks create money

Stage	Currency in circulation	Checkable bank deposits	Money supply
First stage Cash held by the public	$1,000	$0	$1,000
Second stage First deposit → $900 loan and payment			
Third stage $900 redeposit → $810 new loan			

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%



First bank: $100 required reserves; $900 available to lend.

How banks create money

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The first bank lends $900 in cash.

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The borrower pays the same $900 to the supplier.

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Third stage $900 redeposit → $810 new loan	$810	$1,900	$2,710

The next bank receives $900, keeps $90 and lends $810.

The third stage includes the **next $810 loan**.

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

Chapter 14, Part 2 · Section 2

Calculate and interpret the simple money multiplier.

Determine total deposit expansion and explain the assumptions that limit the model’s application to actual banking.

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

rr = 10% · Each row follows lending and payment to the next bank.

Bank	Deposits	Reserves kept	New loan
A	$1,000		
B			
C			
…	…	…	…

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

At Bank A: $1,000 − (0.10 × $1,000) = **$900**.

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Bank	Deposits	Reserves kept	New loan
A	$1,000	$100	$900
B	$900	$90	$810
C	$810	$81	$729
…	…	…	…

The Simple Money Multiplier

Initial cash deposit: \$1,000 · rr = 0.10

$$D = 1{,}000 + 1{,}000(1 - rr) + 1{,}000(1 - rr)^2 + \ldots$$

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$10,000 total deposits includes the original $1,000.

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Under these assumptions, \$1 of reserves supports \$10 of deposits.

Why Reality Is More Complicated

Full redeposit. People deposit the payments they receive.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

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Only required reserves are retained. Banks keep rr × deposits and lend the remaining amount.

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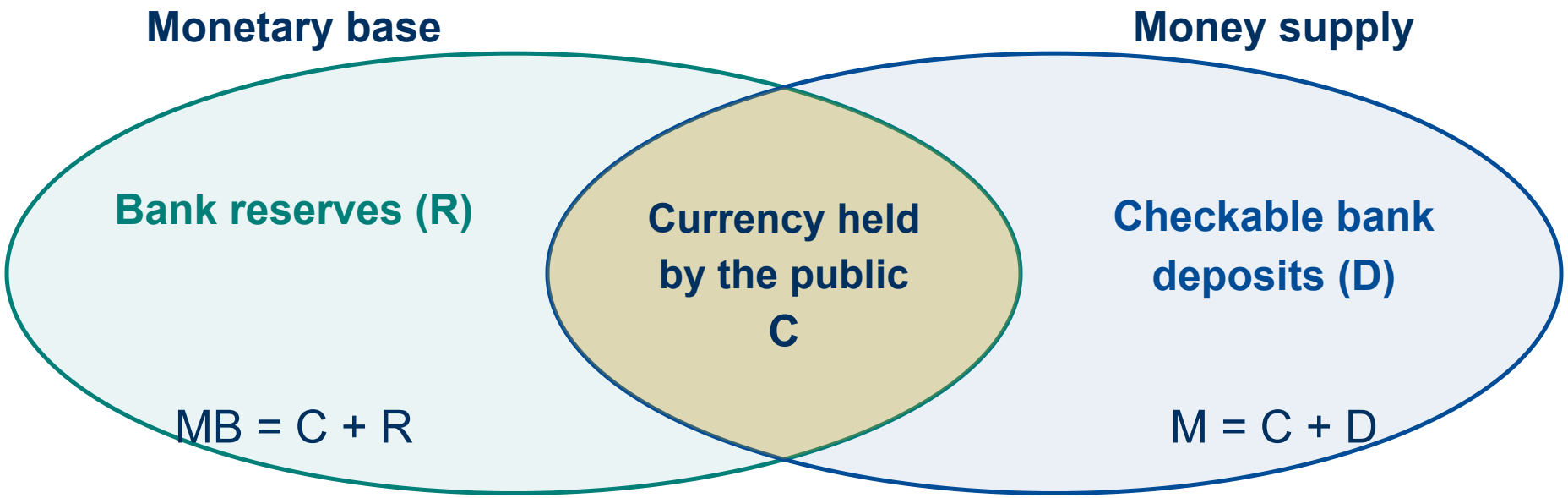
More cash held by the public or more excess reserves
means less deposit expansion.

1/rr is the model's simple deposit multiplier.

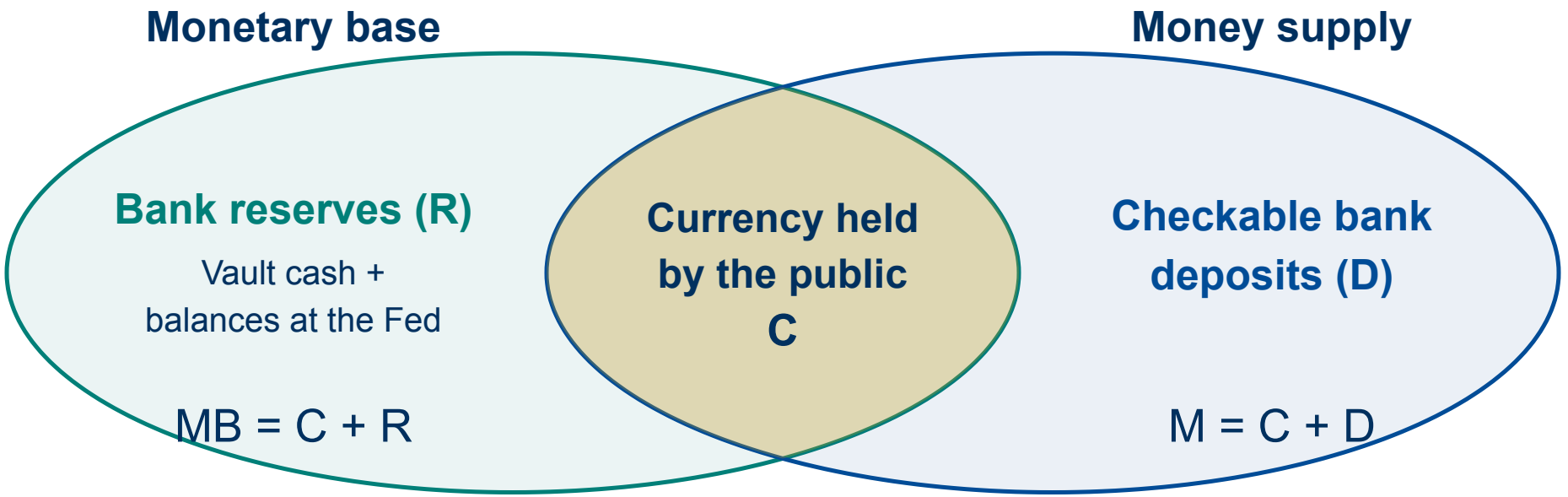
rr = 10% is our example; the U.S. statutory requirement is 0%.

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Monetary Base and Money Supply

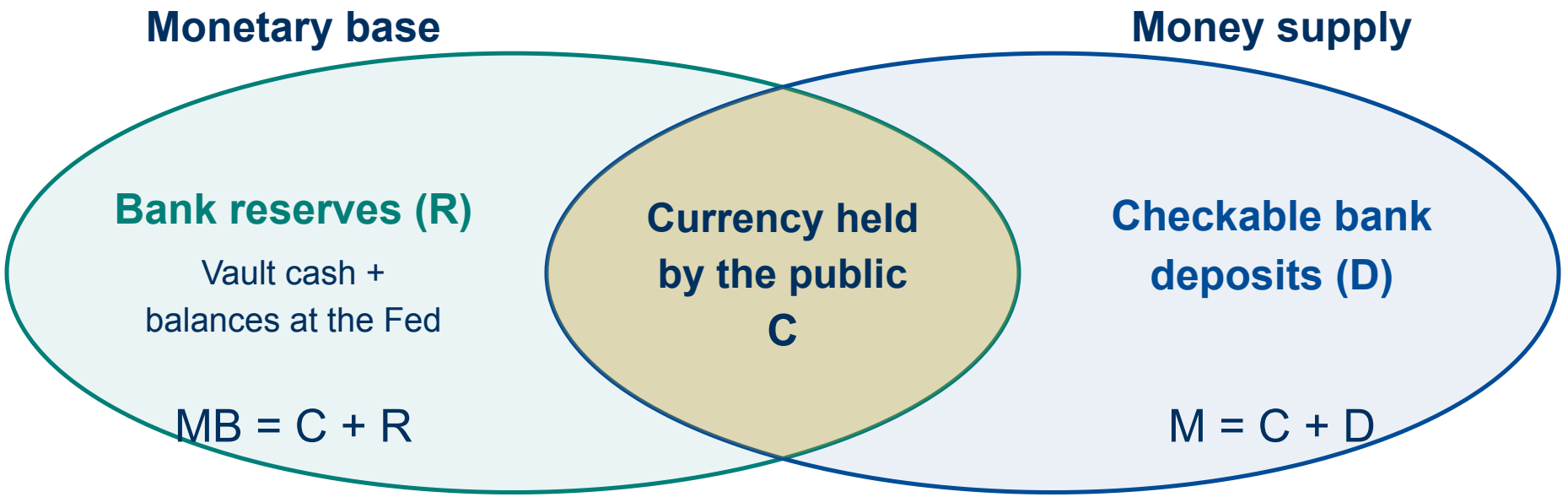


Monetary Base and Money Supply



Public currency is the overlap. Bank reserves and public deposits are different components.

Monetary Base and Money Supply



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$$\text{Money multiplier} = \frac{M}{MB}$$

Next: how Fed operations change reserves and the monetary base.

Money Creation Checkpoint

$1,000 of existing public cash · rr = 10% · Full lending and redeposit

1. Cash becomes a deposit.

Does the money supply rise immediately?

2. Lending repeats across banks.

What are total deposits and the net increase in money?

3. Banks make loans.

Does lending itself create monetary base?

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Does lending itself create monetary base?

No. It creates deposits; it does not create base money.

Chapter 14, Part 2 · Section 3

Understand how central banks work and identify their policy tools.

Explain the roles of the discount rate, interest on reserve balances and open-market operations.

The Federal Reserve

The Federal Reserve (the Fed) is the U.S. central bank.

Its monetary-policy goals: maximum employment and stable prices.

Federal Reserve, System Structure

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The tools we will study:

- Loans from the Fed to banks
- Interest paid on banks' balances at the Fed
- Purchases and sales of bonds

Federal Reserve, System Structure

The Federal Reserve System



1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

12 Reserve Banks. San Francisco also serves Alaska and Hawaii.

Board of Governors

7 governors · Washington, D.C.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

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Board of Governors

7 governors · Washington, D.C.

Each governor:
The U.S. President nominates;
the U.S. Senate confirms.

Full term: 14 years.
One term ends every two years.

Chair: chosen from the governors by the same process, for four years.

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Why regional banks? Share responsibility and bring local economic conditions into national decisions.

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The Federal Open Market Committee

FOMC = the Fed’s monetary-policy committee.

7	+	1	+	4
Board governor seats		New York Fed president		Other regional presidents
Permanent votes		Permanent vote		One-year voting turns

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Directs bond purchases and changes in the Fed's bond holdings.

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For our activity: the statement explains the decision and goals; the implementation note gives the operating steps.

The Board of Governors sets reserve interest and approves discount rates.

Federal Reserve, FOMC

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

New York Fed, Monetary Policy Implementation

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Monetary policy · financial stability and supervision · payments

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FFR: the federal funds rate—the interest rate in this market.

New York Fed, Monetary Policy Implementation

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

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FFR: the federal funds rate—the interest rate in this market.

Policy tool	What changes
Discount rate	Cost of borrowing from the Fed
Interest on reserve balances	Return on eligible balances at the Fed
Open-market operations	Securities and bank reserve balances

Next week: interest rates, spending, and aggregate demand.

New York Fed, Monetary Policy Implementation

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

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Lower discount rate:
Cheaper Fed funding; borrowing becomes more attractive.

Higher discount rate:
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A reservation rate:
a safe alternative return to lending.

Higher IORB raises that benchmark; lower IORB reduces it.

IORB influences lending rates; it is not a fixed minimum rate on every loan.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

Open-Market Purchases

The Fed buys $1,000 of government bonds from a commercial bank.



Purchase: reserves enter the bank.

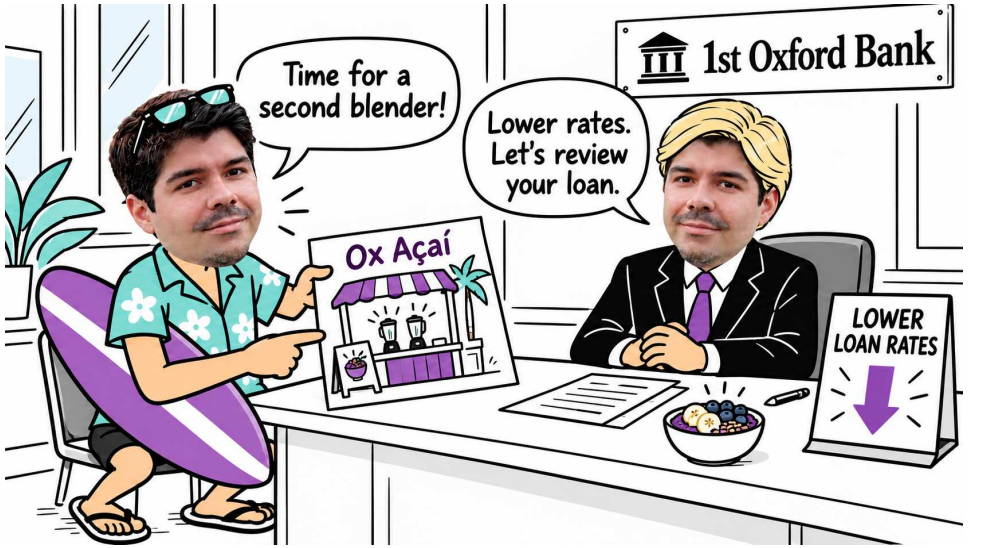
New York Fed, Monetary Policy Implementation

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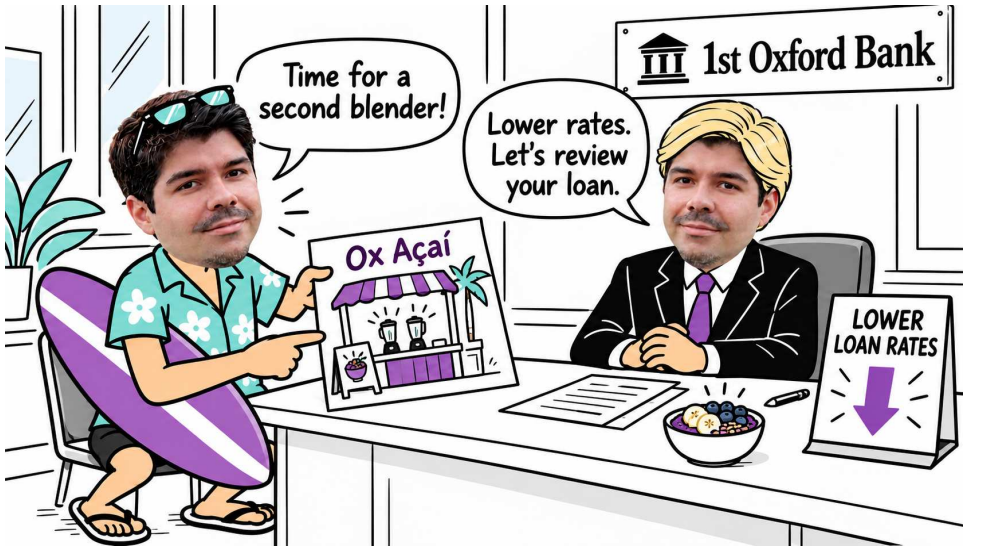
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Bank reserves +$1,000 · Monetary base +$1,000

New York Fed, Monetary Policy Implementation

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Sale: reserves leave the bank.

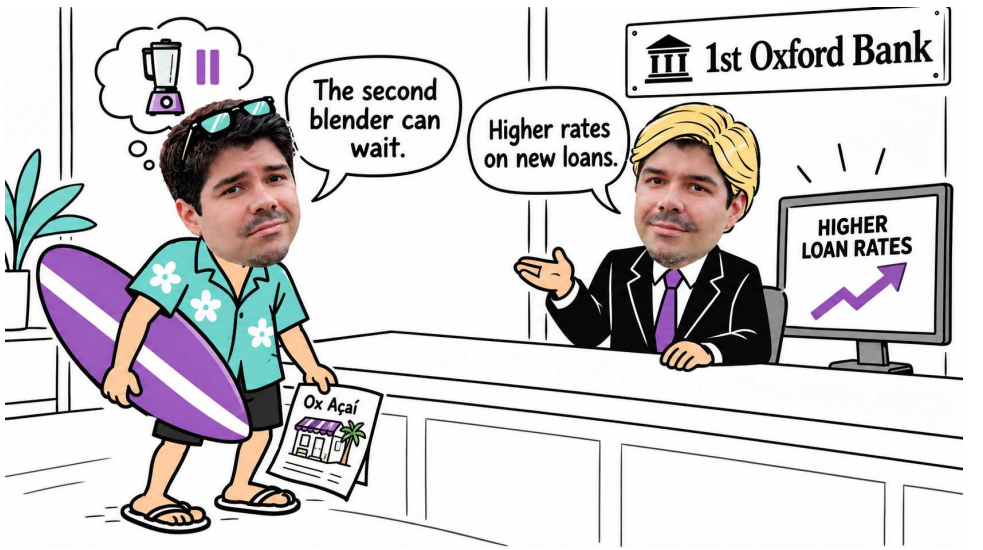
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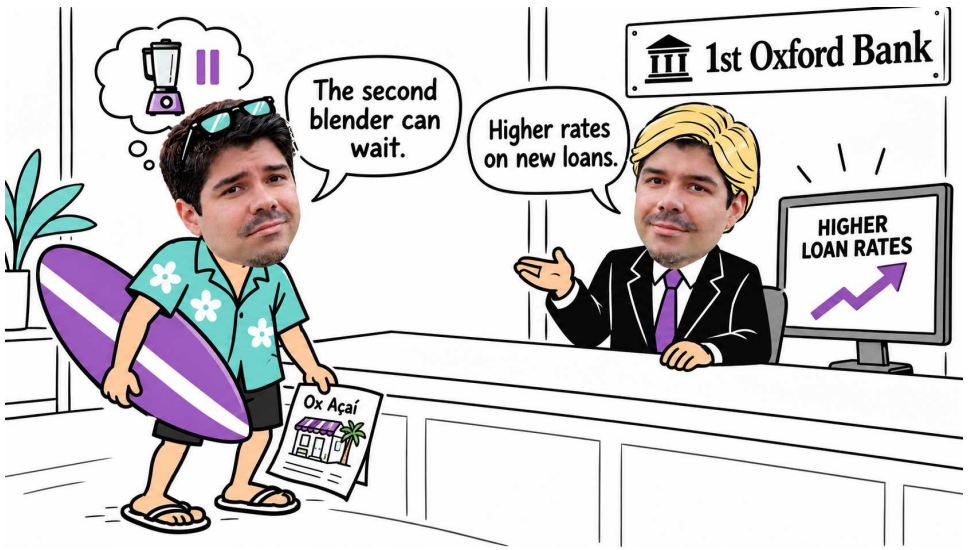
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New York Fed, Monetary Policy Implementation

The ECB and the Federal Reserve

ECB = European Central Bank · Eurosystem = ECB + euro-area national central banks.

	Federal Reserve	ECB / Eurosystem
Central board	7 Board governor seats	6 Executive Board members
Policy committee		
Operating network		
Main policy goals		

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

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Main policy goals	Maximum employment; stable prices	Price stability is primary

Both use policy rates, lending facilities, and asset operations.

The ECB’s deposit facility offers a comparable remuneration benchmark.

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

Chapter 14, Part 2 · Section 4

Understand how lessons from financial crises shaped monetary policymaking.

Connect historical crises to changes in central banks’ responsibilities and policy approaches.

Bank Runs and the Creation of the Fed

1907–1913: panic and the Fed's creation

A failed bet on copper-company shares helped trigger panic in 1907. People rushed to withdraw deposits, and banks struggled to find cash. The Fed was created on **December 23, 1913** to help meet these cash needs.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

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1930–1934: Great Depression bank runs

New runs helped force thousands of banks to close. Lending collapsed, and the Fed failed to stop the crisis. Federal deposit insurance was created in **1933** and began in **1934**.

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Policy lesson: The Fed provides emergency bank loans; the FDIC protects eligible deposits.

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Stagflation and High Interest Rates

1970s: stagflation

High inflation came with weak growth and high unemployment. Oil-price shocks made both problems worse.

Fed History: Great Inflation · Fed History: Recession of 1981–82 · BLS: Annual CPI Inflation

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Under Paul Volcker, short-term interest rates rose to nearly **20%**. Borrowing became much more expensive, slowing spending.

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1981–1983: costs and results

A deep recession pushed unemployment to nearly **11%** in 1982. Annual inflation fell from **13.5% in 1980** to **3.2% in 1983**.

Policy lesson: Lower inflation came with weaker spending and serious job losses.

Fed History: Great Inflation · Fed History: Recession of 1981–82 · BLS: Annual CPI Inflation

The 2008 Crisis and the Fed’s Response

The problem: Investment firms and funds relied on short-term borrowing. Home-loan losses made their lenders pull their money out—a run outside ordinary banks.

2007–2008: discount window and emergency loans

Cheaper, longer loans for banks. Separate emergency programs also lent to some investment firms.

Federal Reserve: Emergency Lending, March 2008 · Fed History: Interest on Reserves · Federal Reserve: Bond Purchases, November 2008

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2008–2009: larger open-market purchases

Bought large amounts of longer-term bonds, including bonds backed by home loans, to lower borrowing costs and support spending.

Policy lesson: The crisis brought broader Fed lending, larger bond purchases and a new tool for guiding rates.

Federal Reserve: Emergency Lending, March 2008 · Fed History: Interest on Reserves · Federal Reserve: Bond Purchases, November 2008